Determining Planck's Constant

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Contents

Abstract

This experiment aimed to determine the value of Planck's Constant h by analysing the voltage generated when photons of differing wavelenghts strike a caesium cathode. The value of $(6.550 \pm 0.387) \times 10^{-34}\,\mathrm{J\,H^{-1}}$ was found, which agrees with the defined value of $6.626\,070\,15 \times 10^{-34}\,\mathrm{J\,H^{-1}}$ (CGPM, 2017)[??]. The work function of Caesium was found to be (1.91 ± 0.13) eV, which conforms to the expected value of 1.95 eV.

1 Introduction

1.1 Background

In 1887 physicist Heinrich Hertz noticed that exposing a spark gap to UV light would increase the maximum length of spark that could be achieved for a given voltage (Heinrich Hertz, 1887)[??]. Alexander Stoletov continued to study the effect, discovering that the intensity of the light on the metal was proportional to the induced current. Based off of Max Planck's work, Albert Einstien hypothosised that because energy was quantized, a threshold energy was required before an electron could be released. As a photon's energy is related to it's frequency and wavelength, only wavelenghts above a threshold can create a current in a material. (Glenn Elert, 1998-2026)[??]

1.2 Theory

The formula Planck derived for the energy of a photon was

$$E = hf \tag{1}$$

(Max Planck, 1901)[??]. The energy of photon required to cause a voltage in the material is

$$KE_{\text{Electron, max}} = E - \phi \tag{2}$$

(Douglas C. Giancoli, 2016)[??]where ϕ is the work function for a given material. The minimum frequency required will occur when $KE_{Electron} \to 0$. Therefore

$$hf_0 - \phi = 0$$
$$f_0 = \frac{\phi}{h} \tag{3}$$

When $f > f_0$ the photoelectric effect will occur. For this experiment, the maximum kinetic energy from a photon is measured by determining the voltage required to stop any current generated by the photoelectric tube. As such

$$eV_{\text{Stopping}} = KE_{\text{Max}} = hf - \phi$$

By plotting the energy eV_{Stopping} against hf the slope h can be determined from

$$h = \frac{\Delta E}{\Delta f} \tag{4}$$

The work function can then be taken as

$$\phi = eV_{Stopping} - hf \tag{5}$$

2 Experiment

2.1 Apparatus

- 3B Scientific Planck's Constant Apparatus 1000537 / U10700-230 (3B Scientific, 2024)??
  Within the supplied apparatus is a
  - 1P39 Photoelectric Tube
  - Voltmeter
  - Nano-ammeter
  - LED power supply
- LEDs of wavelenghts
  - 472 nm
  - 505 nm
  - 525 nm
  - 588 nm
  - 611 nm

2.2 Method

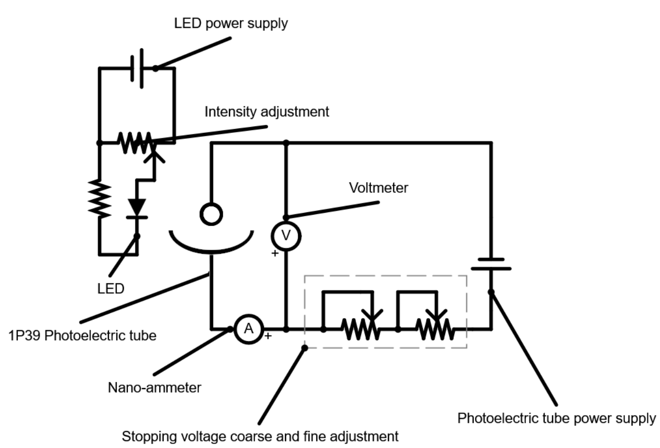


Figure 1: Diagram for the internal circuitry of the apparatus

2.2.1 Part 1

The equipment was set up as shown in figures **??**. The apparatus was connected to mains voltage. The LED's power connector was plugged in and the output secured onto the plastic cover surrounding the photoelectric tube. The intensity was set to 75%. The LED was turned on. The setup was then allowed to stabilize for 1 minute. The coarse adjustment knob was turned until the reading on the nano-ammeter was approximately 0 A. The fine adjustment knob was then turned until the nano-ammeter oscillated between -0 A and $+0$ A, where the critical voltage roughly equaled the back-emf generated by the photoelectric tube. The value on the voltmeter was then recorded. This was repeated for all 5 LEDs.

2.2.2 Part 2

The equipment was set up as shown in figure **??**. The apparatus was connected to mains voltage. The LED's power connector was plugged in and the output secured onto the plastic cover surrounding the photoelectric tube. The intensity was set to 100%. Using the adjustment knobs, the stopping voltage was adjusted until the nano-ammeter oscillated between -0 A and $+0$ A. The voltage was recorded. This was repeated 5 times, each time reducing the intensity by 20%. The LED was then changed for one of a different wavelenght. This was done for the 472 nm, 525 nm and 611 nm LEDs.

3 Results

Name	Symbol	Value	Reference
Electron charge	e	1.602×10^{-19} C	(NIST, 2022)??
Speed of light in a vacuum	c	$2.998 \times 10^{8}\ \mathrm{m\,s^{-1}}$	(BIPM, 2019)??
Accepted value of Planck's constant	h	$6.626 \times 10^{-34}\ \mathrm{J\,H^{-1}}$	(CGPM, 2017)??
Accepted value of the work function of caesium	w	1.95 eV	(David R. Lide, 2008)??

Table 1: Constants and accepted values

3.1 Part 1

Wavelength [nm]	Wavelenght [m]	V_0 [V]	Energies eV_0 [J]	Frequencies [Hz]
472	4.14×10^{-7}	0.663	1.062×10^{-19}	6.36×10^{14}
505	5.05×10^{-7}	0.514	8.23×10^{-20}	5.94×10^{14}
525	5.25×10^{-7}	0.465	7.45×10^{-20}	5.71×10^{14}
588	5.88×10^{-7}	0.161	2.58×10^{-20}	5.10×10^{14}
611	6.11×10^{-7}	0.085	1.36×10^{-20}	4.91×10^{14}

Table 2: Energies and frequencies for the photons

Each of the energies were taken as

$$E = eV_0 \tag{6}$$

Where e is the charge of an electron. The frequencies were taken as

$$f = \frac{c}{\lambda} \tag{7}$$

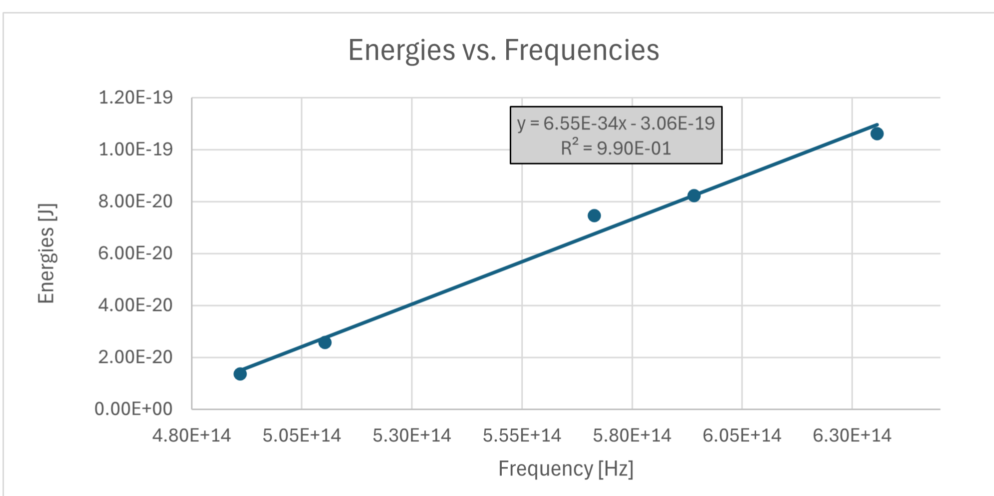


Figure 2: Graph of the energies and frequencies from table **??**

h [JH^{-1}]	6.54×10^{-34}	Work function [J]	-3.06×10^{-19}
Δh [JH^{-1}]	3.9×10^{-35}	ΔWork function [J]	2.17×10^{-20}
r^2	0.989	ΔE [J]	4.6×10^{-21}
F statistic	287.1	Degrees of freedom	3
ssreg	6.102×10^{-39}	ssresid	6.376×10^{-41}

Table 3: LINEST results for table **??**

The Excel command used to generate the LINEST table was

```
LINEST(Energies column, Wavelenghts column, TRUE, TRUE)
```

which is analagous to equation **??**. Converting the work function to units of eV a value of

$$\phi = \frac{3.06 \times 10^{-19} \text{ J}}{1.602 \times 10^{-19} \text{ JeV}^{-1}} = 1.91 \text{ eV}$$

was determined.

3.2 Part 2

Lambda [nm]	472	525	611
Intensity	Voltage [V]	Voltage [V]	Voltage [V]
100	0.658	0.47	0.083
80	0.659	0.469	0.083
60	0.66	0.469	0.083
40	0.66	0.469	0.083
20	0.66	0.469	0.083
0	0.66	0.469	0.083

Table 4: LED voltages at each intensity

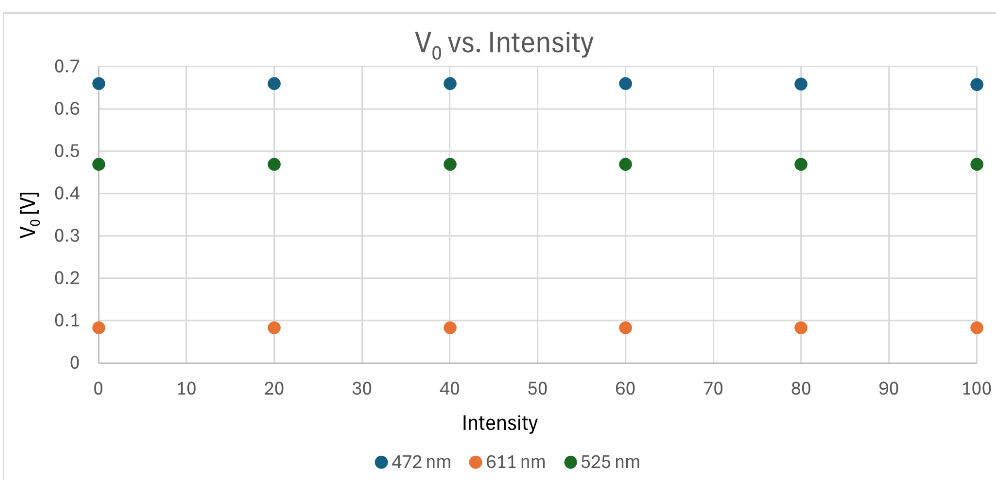


Figure 3: Graph of the V_0 and intensity values from table **??**

As can be seen in figure **??**, the voltages remain constant within experimental error for every intensity. This shows that the stopping voltage does not depend on the light source's intensity and instead only on the wavelenght of the LED.

3.3 Error Analysis

3.3.1 Quantitative

Quoted errors listed in the apparatus manual:

$$\Delta V_0 = 0.5\%$$

$$\Delta\lambda = 1 \text{ nm}$$

$$\Delta A = 1\%$$

(3B Scientific, 2024)[??]. From table **??**:

$$\Delta h = 3.9 \times 10^{-35} \text{ J H}^{-1}$$

$$\Delta\phi = 2.2 \times 10^{-20} \text{ J} = 0.13 \text{ eV}$$

3.3.2 Qualitative

One potential source of error comes from the design of the photoelectric tube. As seen in figures **??** and **??**, the

Figure 4: The 1P39 photoelectric tube inside of the Planck's constant apparatus. (The Value Meuseum, 2013)[??]

photoelectric tube consists of a curved photocathode with a thin anode in front of it. Depending on the format of the LED they can often be modeled as a Lambertian point emmitter (Hongming Yang et al., 2008)[??]. Based off of this the setup can be partially simulated using an optical simulation suite. For this purpose Aether's raytracing tool Selene was used. The ray tracing setup can be seen in figure **??** The cylinder, in white, models the photocathode. Selene is still in development so there is not a way to model a simple, curved surface like the photocathode. To get around this the Lambertian source is placed inside the cylinder. The throw of light is 180°, which cannot be modified. In order to not cause abberations in the later analysis stage the source was placed at the center of curvature for the surface and thus the center of the cylinder. In the experimental setup the source of light is beyond the center of curvature for

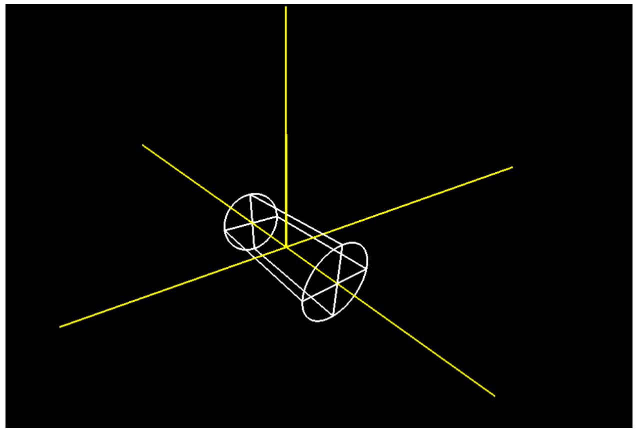


Figure 5: Set up within Selene.

the photocathode, however the argument remains the same. By setting the cylinder to be act as a transparent sensor, information like ray hit distributions on it's surface could be recorded. Then, using Aether's raytracing analysis tool, the distribution of photons across the inside of the cylinder can be found.

Two raytracing simulations were ran. One had the souce intensity set to 1 and the other to 0.5. The results from the two runs are shown in table **??**. The pixel-wise difference between the two runs was taken. An average of a band 10 pixels wide at the center was taken across the image, effectively sweeping across the curved area of the sensor. The results are shown in figure **??**.

From figure 6.1 the difference in deposited power closely follows a bell curve distribution. The difference being small at either edge of the photocathode could have adversily the amperage readings.

Another source of error came from the temperature and movement sensitivity of the apparatus. The stopping voltage changed slightly if the apparatus was touched for too long, left near direct sunlight or if the cable was slightly moved.

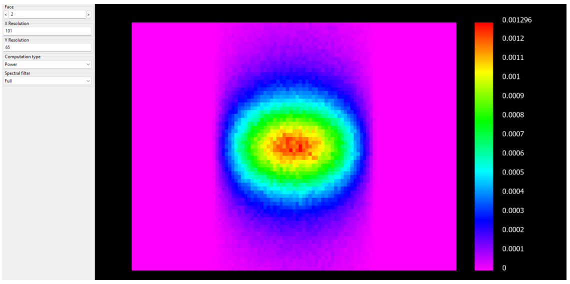
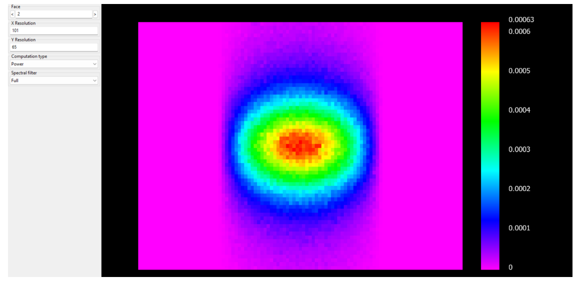
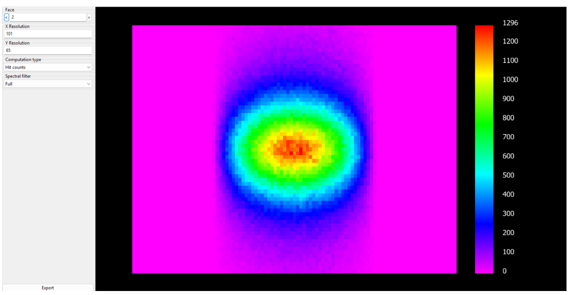
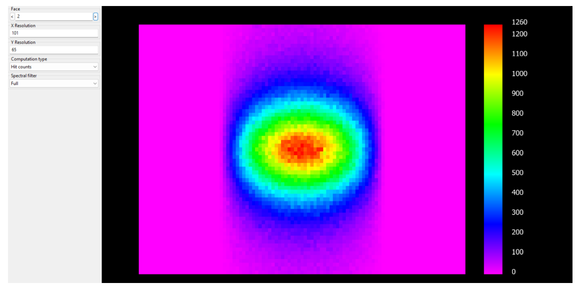
Full power run	Half power run
	
Power distributions	
	
Hit distributions	

Table 5: Recorded power and hit distributions for the two runs

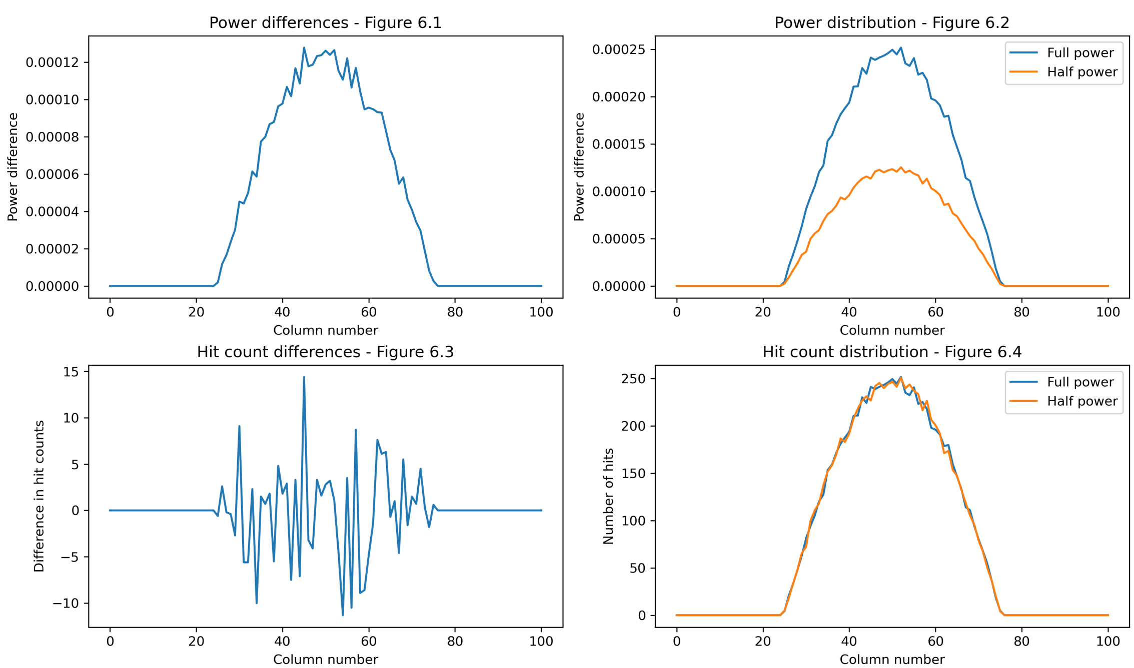


Figure 6: Full power results - Half power results

4 Discussion

The design of the photoelectric tube wasn't optimal for this experiment. The spread of the photons, likely adversily affected the results. This error could have been reduced by collimating the LED light.

Additionally, by placing the apparatus in a temperature controlled enviornment any errors caused by temperature fluctuations could have been negated. To prevent the LED accidentally being moved the intensity controls could have been moved further away.

Depending on the source, the accepted value of caesium differed. Some, like The Handbook for Physics and Chemistry **??** listed it as being 1.95 eV, while the apparatus's manual **??** listed it as being approximately 2 eV.

5 Conclusion

By analysing the stopping voltage required to prevent current flow for a given LED the value of Planck's constant could be determined. The value of Planck's constant was found to be $(6.54 \pm 0.39) \times 10^{-34}\ \mathrm{J\,H^{-1}}$, which conforms to the defined value of $6.626\,070\,15 \times 10^{-34}\ \mathrm{J\,H^{-1}}$ (CGPM, 2017)**??**. The work function was determined to be (1.91 ± 0.13) eV, which agrees to the acccepted value of 1.95 eV. The intensity was found to have no link to the stopping voltage. The experiment could've been made more accurate following the points made in section 4.

Notes

This document was compiled using LaTeX. The source files, including images and the python script used to generate figure **??**, for this document are available at (https://github.com/daireobrien61/Formal-Report-Experiment-21). Websites like Overleaf (www.overleaf.com/learn), The TEX Stackexchange (tex.stackexchange.com) and each package's documentation on their respective CTAN (www.ctan.org) pages were referenced heavily for good practice and specific commands. Where possible the links to specific answers have been included in the source code as comments. The circuit diagram was made using Paul Falstad's circuit simulator (www.falstad.com/circuit). The ray tracing simulations were done using the Aether optics simulation suite (aether.utt.fr), specifically their ray tracer Selene and analysis programs.

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